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General Comment

I work in a physical science field and have been part of discussions about the use of AI in my field for the past 6-7 years. I have watched it grow from a new concept in my field to something that is regularly used. I know the technology has a lot of potential uses, and I also know that relying on it too much and too quickly is imprudent and potentially dangerous.

AI systems are only as good as how they are trained. I think a lot of people view AI as a solution to a lot of problems, but they forget or fail to realize that it's incredibly difficult (or potentially impossible) to train AI/ML with a dataset that contains all possible scenarios or outcomes; it's arguable that most or all information in a training dataset will represent something within the bulk of a statistical distribution, but may fail to capture information about instances that exist in the tails of the distribution. It's these instances that are often the most impactful on humans or human society, because they are so rare. This may be something like a powerful earthquake, a very strong storm, a significant societal event... if similar occurrences of these rare events are not in the AI's training dataset, how is the AI system supposed to accurately anticipate them?

It's for this reason that humans must also continually oversee AI/ML operation; the AI system does not operate considering any rules or physical laws or anything else, and it can put forth solutions that are nonsensical or unlikely. Human oversight is required as a means to quality control AI output. This is something that might be easily overlooked if we begin to rely too much on AI; we risk brain drain (e.g. losing our understanding of the processes we are using AI to represent), and we could begin to fail to catch instances where AI is providing obviously wrong answers. This could be devastating; an obvious scenario like this would be a passenger in a self driving car failing to hit the brakes when the AI operating the car can't identify a road hazard. The idea that we don't need humans if we have AI is false.

In addition to needing good training and continual oversight from humans, AI systems also often need to be continually updated or re-trained. We live in a constantly evolving society on a constantly evolving planet. What is "normal" today may not have been "normal" 5 years ago, and will likely not be "normal" 5 years in the future. AI/ML systems, like most other models, need frequent updating and maintenance. They can't just be set up, left to do their thing, and then be expected to continually work.

There are likely many more reasons why AI needs to be treated carefully and with much consideration. It would be ill advised to move too quickly to completely abandon the non-AI methods of doing things that have served us well for years in favor of AI-only solutions. AI is a tool, and it should most often be used in conjunction with other methods for accomplishing goals. If we over-rely on it, we risk throwing the proverbial baby out with the bathwater. And if we do this to replace human labor, we also risk causing widespread societal unrest. The human population continues to grow exponentially, and we continue to exist in a society where one must pay money to be alive. No one asks to be born, and the more we replace human jobs with computers and machines, the more we will see human beings struggle to earn the money they need to pay to live the lives they never asked for.